

Exploration of a Text Corpus: Feedback from the Thriving in ECS Workshop

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Introduction:

In Spring 2024, the College of Engineering and Computer Science (ECS) piloted a mandatory academic recovery workshop titled “Powering Through Academic Recovery” for students on the highest level of probation whose suspension appeals were approved. This new model replaced the formal appeal process with a required intervention that aimed to provide students with early support and a space for reflection. The workshop, collaboratively facilitated by advisors from ECS Career Services and the Student Success Advising team, was held on January 26, 2024. The focus was on experiential learning activities grounded in design thinking. Key themes included mind mapping, problem framing, prototyping, and articulating the NACE (National Association of Colleges and Employers) career competencies.

This project uses student survey responses from before and after the workshop to build and explore a text corpus. The goal is to apply text mining techniques to uncover patterns in student sentiment, reflection, and engagement while preparing the data for potential future classification or clustering models.

Data and Exploratory Data Analysis (EDA):

The data for this study was extracted from two structured feedback forms: a pre-workshop and post-workshop survey. Each form collected open-ended responses from student participants, along with their assigned success advisor and any lingering concerns. After cleaning and filtering unnecessary metadata, each student’s response was treated as an individual document. The final dataset contained 26 documents, 13 labeled “Pre” and 13 labeled “Post.” This label marks the time of the response relative to the workshop.

The constructed corpus was saved in a .csv format with the following columns:

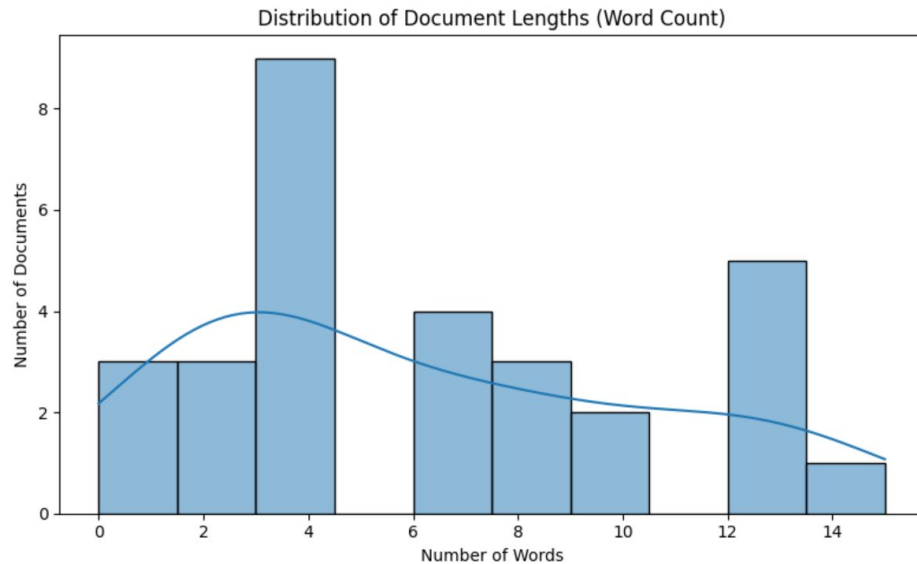
- DocID: A unique identifier for each document
- Label: Pre or Post
- Text: Combined cleaned responses

Each document represents the voice of a student preparing for or reflecting on the workshop, offering insights into academic attitudes, self-awareness, and growth.

Exploratory Data Analysis:

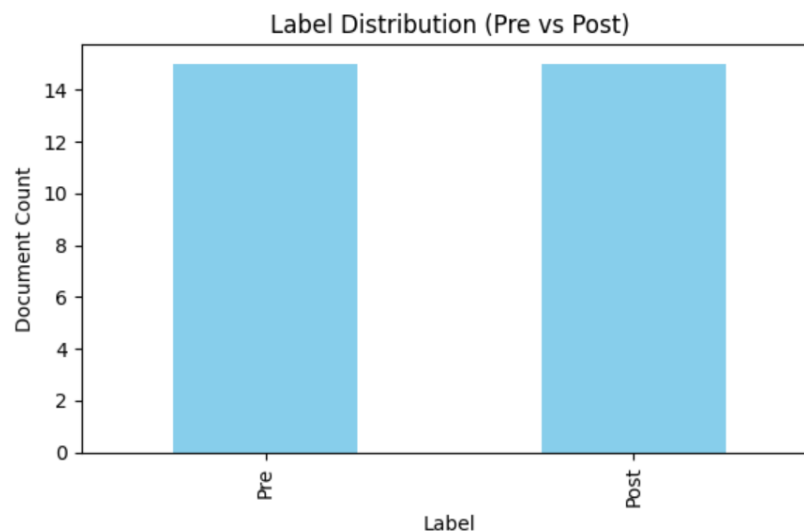
To deepen our understanding of the text data, an exploratory data analysis (EDA) was conducted. This included inspecting document lengths, word frequencies, and label distributions. English stopwords were removed using the NLTK corpus, and a customized stopword list was created to eliminate additional common but uninformative words that appeared frequently in feedback (e.g., 'workshop', 'session').

Document Length Analysis:



A histogram of document lengths revealed that most student reflections were relatively short, with most responses clustered around 4–5 words. This right-skewed distribution shows that while a majority of students submitted brief responses, a few provided more detailed feedback (up to 15 words). This variation could reflect differences in engagement or comfort with open-ended questions.

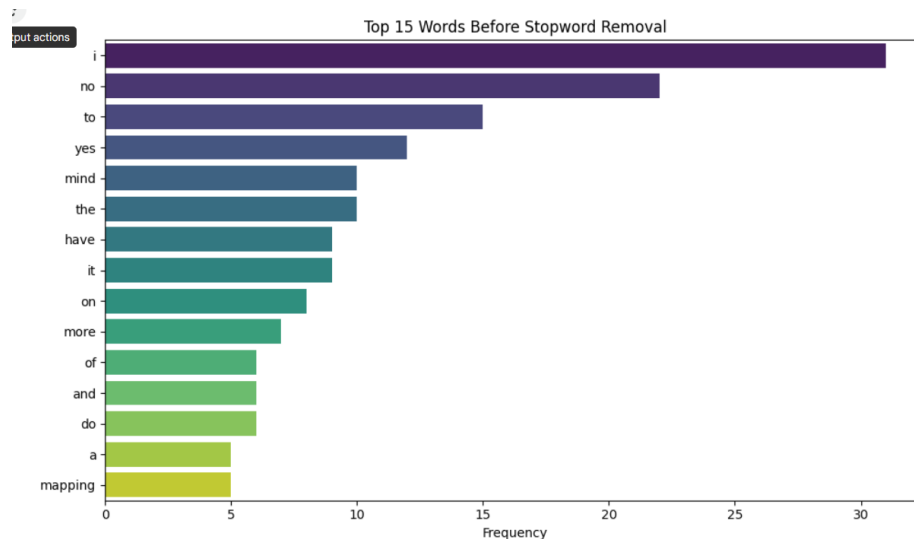
Label Distribution:



The label distribution chart confirms an equal number of student reflections collected before and after the academic recovery workshop, with 15 “Pre” and 15 “Post” documents. This balance strengthens the validity of comparative analyses and makes sure that no group dominates in future analysis. Maintaining an even label distribution helps reduce model bias and allows

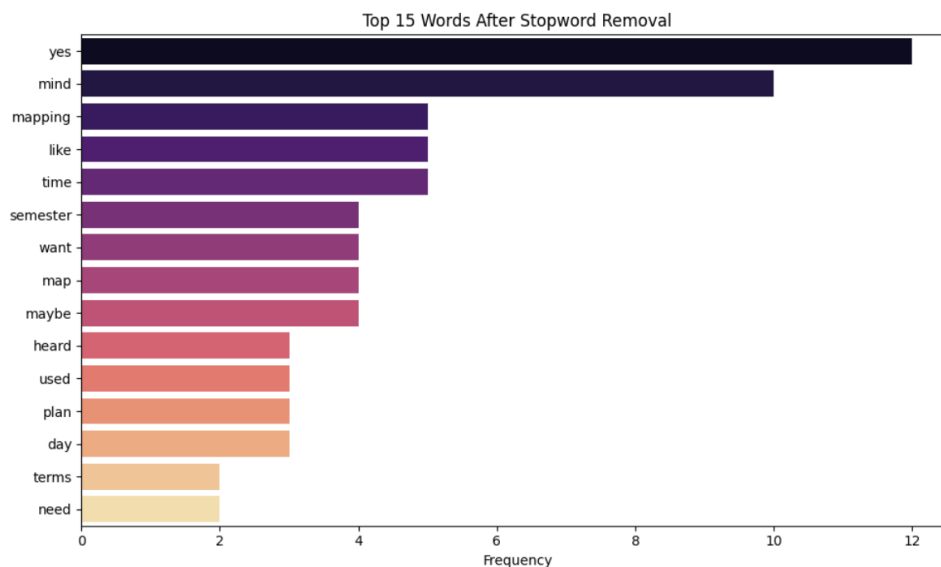
clearer interpretation of any observed differences in student sentiment or language patterns between the two groups.

Term Frequency Before Stopword Removal:



A bar plot of the top 15 most frequent words in the raw corpus revealed that most common terms were stopwords, such as “i”, “no”, “to”, “the”, and “and.” These words can occur often in English but have little analytic value when interpreting student feedback. This justified the decision to apply both the NLTK standard stopwords list and a domain-specific extension which included words like “workshop”, “session”, and “advisor.” Removing these terms helped focus the analysis on semantically meaningful content.

Term Frequency After Stopword Removal:



Methods and Models:

After cleaning and lemmatizing the text, the documents were vectorized using the TF-IDF (Term Frequency–Inverse Document Frequency) method, which is a widely used technique in text mining that highlights words that are important in individual documents while down-weighting terms common across the entire corpus (Ramos, 2003). Both single words (unigrams) and two-word phrases (bigrams) were included, and the vocabulary size was capped at 1,000 terms to prevent overfitting and manage sparsity in the small dataset.

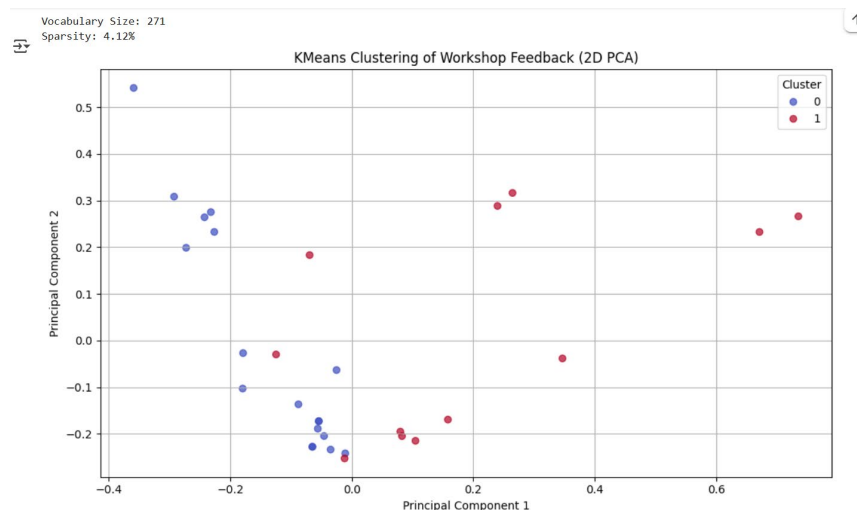
To explore whether patterns in student feedback aligned with the timing of the response (Pre vs. Post), a KMeans clustering model was applied with $k = 2$ clusters. Although KMeans is an unsupervised learning method and does not use labels during training, the resulting clusters were compared to the actual Pre and Post categories. This comparison allowed for an examination of how students' language use may have changed over time, helping to assess the evolution of textual themes resulting from the workshop experience (Jain, 2010).

To complement the clustering and lexical analysis, sentiment analysis was performed using the VADER (Valence Aware Dictionary and sEntiment Reasoner) tool from the NLTK library. VADER is specifically designed for evaluating short, informal text such as social media posts and survey responses which made it a strong choice for analyzing student feedback (Hutto & Gilbert, 2014). Each cleaned document was scored along four dimensions: positive, negative, neutral, and compound sentiment. The compound score, a normalized metric ranging from -1 (most negative) to $+1$ (most positive), was used to summarize overall sentiment.

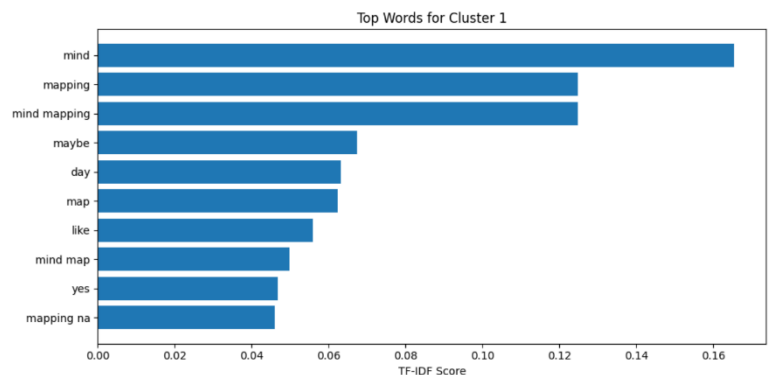
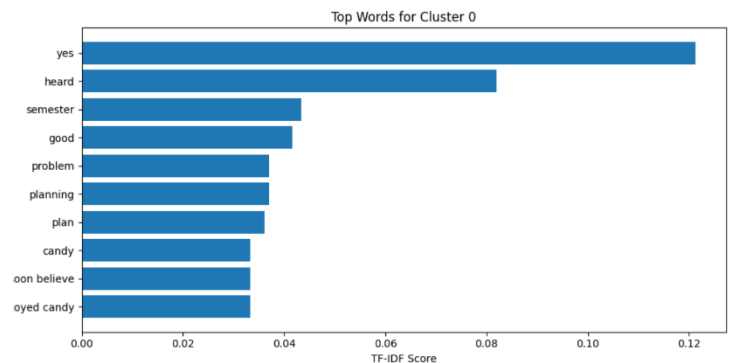
Results and Observations:

To explore how student reflections shifted from before to after the workshop, a combination of clustering, word frequency analysis, and sentiment analysis was used. These methods helped identify patterns in language, emotional tone, and overall focus in the students' written feedback.

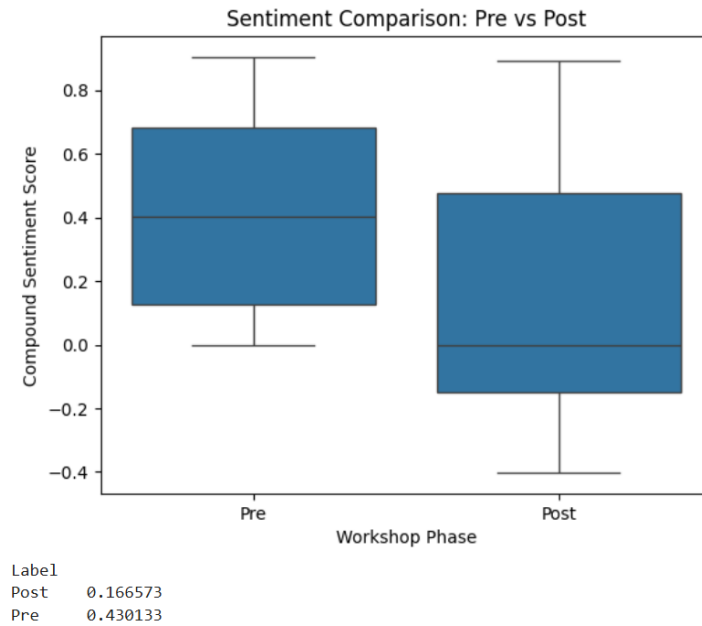
First, a KMeans clustering model was applied to the TF-IDF matrix of the feedback texts to determine whether natural groupings aligned with the Pre and Post workshop phases. Although KMeans does not use labels during training, the resulting clusters showed a strong visual separation when reduced to two dimensions using Principal Component Analysis (PCA). As shown in the PCA scatterplot, Cluster 0 and Cluster 1 mostly aligned with Pre and Post feedback. This suggests that students' language patterns shifted enough to be distinguishable without requiring labels, indicating that the workshop likely had an impact on how students expressed their thoughts.



Clustering results helped distinguish patterns in how students described their experience before and after the workshop. Cluster 0 included more Pre-workshop responses and was characterized by terms like “problem,” “help,” “time,” and “plan”, which suggest concern and a desire for academic improvement. Cluster 1 included mostly Post-workshop responses, which featured words like “mind map,” “enjoy,” and “connections,” pointed to a shift in tone toward reflection and engagement. Word clouds and bar charts below made these differences easy to see, and supported the evolving student mindset throughout the intervention.



To further explore emotional tone, a sentiment analysis was conducted using VADER (Valence Aware Dictionary and sEntiment Reasoner). Interestingly, Pre-workshop responses showed slightly more positive compound sentiment scores on average. Many students wrote hopefully about what they wanted to gain or improve. Post-workshop responses, while still positive, tended to be more balanced or neutral, focusing on what they learned or experienced. A boxplot comparing compound scores showed that Pre responses had a wider and higher range of sentiment values than Post responses, which suggests that students began the workshop with strong aspirations and ended with a more grounded view of their academic path forward.



Conclusion:

This project explored how student feedback from an academic recovery workshop could be cleaned, organized, and analyzed using text mining techniques. By applying lemmatization, custom stopwords removal, and TF-IDF vectorization, we were able to make sense of short, informal survey responses. Visuals like word clouds and bar charts helped highlight changes in language before and after the workshop.

Even though the clustering model didn't use the Pre or Post labels during training, the two clusters showed clear patterns. Pre-workshop feedback often focused on planning, problems, and uncertainty. In contrast, Post-workshop responses reflected more engagement and self-awareness. Sentiment analysis supported these findings, showing that tone and emotion also shifted after the workshop.

Overall, the results suggest that the workshop had a positive effect. Students moved from optimistic uncertainty to more thoughtful and confident reflections. This project shows how text mining can be a useful way to understand student experiences and is a strong starting point for more advanced modeling in the future.

References:

- Hutto, C. J., & Gilbert, E. (2014). *VADER: A parsimonious rule-based model for sentiment analysis of social media text*. In *Proceedings of the International AAAI Conference on Web and Social Media* (Vol. 8, No. 1).
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